

Final Task - Aircraft Model

Task

You need to develop a program (in any programming language) for the Research Civil Aircraft Model (Class 12 – Research Aircraft Civil Model) where:

1. (1) Implement an `xdot` function in which all intermediate steps are clearly developed and well documented, with the following inputs and outputs:

```
def xdot(X,U):  
    # State and Control vector  
    x1 = X[0] # u  
    x2 = X[1] # v  
    x3 = X[2] # w  
    x4 = X[3] # p  
    x5 = X[4] # q  
    x6 = X[5] # r  
    x7 = X[6] # phi  
    x8 = X[7] # theta  
    x9 = X[8] # psi  
  
    u1 = U[0] # aileron  
    u2 = U[1] # stabilizer  
    u3 = U[2] # rudder  
    u4 = U[3] # throttle 1  
    u5 = U[4] # throttle 2  
  
    # Constant definitions  
  
    # Step 1  
  
    # Step 2  
  
    # Step 3
```

```

# Step 4

# Step 5

# Step 6

# Step 7

# Step 8

# Step 9

# Step 10

return X_dot # vector state derivatives

```

2. (1) Using a discrete time step of 1 second during 3 minutes (3×60 s), calculate the integration of the RCAM output and create a loop to simulate the aircraft dynamics with the following initial state in SI units:

$$X_0 = [85 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0.1 \ 0]^T$$

$$u_0 = [0 \ -0.1 \ 0 \ 0.08 \ 0.08]^T$$

- a. Present the behavior of the aircraft in a 3D or 2D graphic interface and plots of all state variables as a function of time, and explain the observed behavior of the aircraft
3. (1) Create a function that deflects the ailerons by +5 degrees for 2 seconds at $t = 30$ s, and:
 - a. Present the behavior of the aircraft in plots of all state variables as a function of time, and explain the observed behavior of the aircraft
4. (1) Create a function and/or modify the code to simulate an engine shutdown in one state, and:

- a. Present the behavior of the aircraft in plots of all state variables as a function of time, and explain the observed behavior of the aircraft
- b. (1) Find the equilibrium point using a swarm optimization algorithm for steady, straight-and-level, trimmed flight at 78 m/s, flying to northeast direction $\psi = 45 \text{ deg}$
- c. Define and justify your cost function.

Tip: you need to identify all constraint to reach steady state straight and level trimmed flight and then create a function of all constraints with the following way:
- d. Explain the setup of you optimization algorithm (number of particles, iteration, etc)
- e. State and control variables to reach trim sate

Deliverables

1. Code (well commented) in a [github](#) repository, check the following [tutorial](#) to upload your code and add the link in the report.
2. Short report (Max 5 pages) explaining:
 - a. Your backend (matrices, equations, calculations, modules, etc.)
 - b. Key observations from your test cases.
 - c. Plots of your state variables
 - d. Cost function explication

Guidelines

- Deadline: May 20 - 8:00 am
- You can use LLMs (Large Language Models) but you must understand all your code. The professor is going to ask you about an random section of your code. If you don't answer with criteria, you will only reach half of the score of the task.
- The maximum number of students are three (No exceptions)

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